Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Python · Full-Stack · PropTech · Affordable Housing Compliance · AWS · Microservices · Complex Data Modeling

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PROFESSIONAL SUMMARY

Senior software engineer with 8 years designing and owning **complex compliance data models for affordable housing and mixed-income portfolios** at Entrata — a direct AppFolio competitor serving the same property managers. Built LIHTC (Tax Credit) set-aside enforcement (20/50, 40/60), AMI threshold calculations, HUD Section 8/HCV HAP reconciliation, annual/interim recertifications (HUD 50059), income eligibility verification, and mixed-income portfolio tracking across **100K+ units** with a deterministic rules engine and full audit trail — **zero compliance violations**. I designed the compliance data models this team now maintains. Owned **20+ production modules end-to-end** (1M+ records, 100K+ MAU, 99.5% uptime) across 120+ Agile sprints; led three platform-level initiatives: Docker microservices rollout (50% speedup), AWS S3/RDS/Glue migration (100% PHI compliance), and first production AI pipeline on **1.6M real financial records** with quality gates and HITL review. Mentored 5+ engineers, 300+ code reviews. Stack: Python, PHP, JavaScript, PostgreSQL, AWS — Ruby fast ramp from OOP/dynamic-typed foundation. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University
Aug 2024 – Aug 2025

Research Associate — ML Pipelines · Anomaly Detection · Statistical Modeling

- Built end-to-end ML pipelines (Python, R, 190+ countries) achieving **$R^2 = 0.977$**; applied Isolation Forest anomaly detection across 3 research pipelines; automated statistical reporting driving **40% improvement** in stakeholder comprehension.

Entrata
Jan 2021 – Aug 2024

Senior Software Engineer — Affordable Housing & Mixed Portfolio Compliance · Full-Stack · AWS · Data Modeling

- Designed and owned **complex compliance data models for mixed-income portfolios** end-to-end — LIHTC (Tax Credit) set-aside enforcement (20/50, 40/60), AMI threshold calculations, annual/interim recertifications (HUD 50059), income eligibility verification, and mixed-income portfolio tracking across **100K+ units**; deterministic rules engine with full audit trails — zero compliance violations.
- Led **20+ production modules** (1M+ records, 100K+ MAU, 99.5% uptime) across 120+ Agile sprints; drove **three platform-level initiatives**: Docker microservices rollout (50% execution speedup), AWS S3/RDS/Glue migration (100% PHI compliance, zero incidents), and first production AI pipeline on **1.6M real financial records** (PyTorch, OpenAI API) with quality gates and HITL review.
- Mentored **5+ engineers**, conducted **300+ peer code reviews**; drove architectural improvements across 200+ SQL queries (30% performance gain); advocated for and shipped platform-wide standards — containerization, cloud migration, AI pipeline — across teams.

Entrata
Jan 2019 – Dec 2020

Software Engineer — HUD/Section 8 Compliance · Data Pipelines · API Integrations

- Built HUD Section 8/HCV compliance modules: HAP reconciliation, household composition tracking, student rule enforcement, asset imputed income calculations; built **API integrations** across property management systems; 300+ peer code reviews; prototype through production in Agile sprints.

Entrata
Jan 2017 – Dec 2018

Associate Engineer — Full-Stack · Python · PHP · PostgreSQL · REST APIs

- Built and shipped full-stack features and SQL data layers (PHP, JavaScript, PostgreSQL) for property management workflows; 0→1 prototyping through production rollout across 100K+ unit portfolios.

PROJECTS

Leasel: Agentic Affordable Housing Compliance System
PROPTECH AGENTIC RAG

Python · LangGraph · ChromaDB · Deterministic Rules Engine · LLM-as-Judge — github.com/venkatalakshmiparimala/Lease_complaince_raq

- **93.6% precision, 87.9% F1** on 1,240 residential lease PDFs across LIHTC/Section 42 compliance rules — agentic LangGraph RAG pipeline, deterministic rules engine for fully auditable compliance determinations; pass@k eval harness (pass@1 81.7%, pass@3 88.3%) with LLM-as-judge scoring.

AgentProbe: Multi-Agent AI Eval Framework
MULTI-AGENT LLM-AS-JUDGE

Python · LangGraph · FastAPI · ChromaDB · React · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- 3-agent LangGraph runtime with tool calling, inter-agent state handoffs, **5-metric LLM-as-judge eval engine**, CI/CD regression gates, 15-endpoint FastAPI backend, and **React observability dashboard** — blocking failing agent runs before production.

AI on FHIR: HIPAA-Compliant Medical Records NLP Platform
HEALTHCARE

Python · Flask · React · FHIR APIs · BERT · LLMs · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- ~60% reduction in clinical retrieval time — NLP platform translating natural-language queries into FHIR API calls via LLMs + BERT; HIPAA-compliant RBAC, end-to-end from design to deployment.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — May 2026 · GPA 3.87

ML systems, statistical modeling, data engineering, AI pipelines, system design, data governance

B.Tech Computer Science & Engineering — JNTU Anantapur — 2013 – 2017

TECHNICAL SKILLS

Languages & Backend: Python, JavaScript, TypeScript, PHP, SQL, PL/SQL, R; FastAPI, Django, Flask, REST APIs, webhooks, RabbitMQ, SOA, microservices, event-driven systems; Ruby (fast ramp — OOP/dynamic-typed foundation)

Data & Compliance: Complex data model design, compliance rules engines, audit trails, LIHTC/HUD/HCV, income eligibility verification, AMI threshold calculations; PostgreSQL, MySQL, query optimization, data pipelines, ETL; React, HTML/CSS

AI & Infrastructure: LangGraph, LangChain, RAG Pipelines, LLM-as-Judge, OpenAI API, PyTorch, BERT, ChromaDB; AWS (S3, EC2, RDS, Glue), Docker, CI/CD, GitHub Actions, TDD, Agile/Scrum, HIPAA/PHI compliance, Git, Linux